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Thomson coil design and potting process

Abstract

An assembly for manufacturing the insulation layer of a Thomson coil includes a base plate, a coil housing, and a cover plate. The base plate securely seats the coil housing, and the coil housing securely seats a Thomson coil and the cover plate. The cover plate has several ribs that hold multiple turns of a Thomson coil in place while epoxy is applied to the coil, thus ensuring that the epoxy is evenly distributed on the coil surface and that the coil windings remain level. The cover plate and coil housing are structured to either receive a high-pressure epoxy injection or to be used in an epoxy potting process, during which all exposed areas of the Thomson coil are coated by liquid epoxy. After the epoxy has solidified, the Thomson coil is coupled to the coil housing, and the housed and insulated Thomson coil is removed from the assembly.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Patent Application Ser. No. 63/398,602, filed Aug. 17, 2022 and entitled, “Thomson Coil Design And Potting Process”.

FIELD OF THE INVENTION
(1) The disclosed concept relates generally to Thomson coils, and in particular, to systems and methods used to produce an insulation layer for Thomson coils.
BACKGROUND OF THE INVENTION
(2) Thomson coil assemblies are used to achieve ultra-fast motion and/or displacement in various applications. One well-known application is the use of Thomson coil assemblies as the actuators in the switching mechanisms of circuit breakers, including hybrid circuit breakers. Oftentimes, a circuit breaker includes one stationary separable contact and one movable contact which need to be in physical contact to conduct power. The stationary separable contact is connected to a stationary

conductor that remains fixed in place, and the movable separable contact is connected to a movable conductor assembly that can be driven between a closed position and an open position by an actuator, such as a Thomson coil assembly. When the separable contacts are closed and the trip unit detects a fault condition, the trip unit transmits a signal to cause the actuator to drive the movable conductor assembly away from the stationary separable contact and to the open position, in order to interrupt the flow of power.

(3) Circuit breakers are designed to interrupt the flow of power quickly, and arcing can occur as a result of opening the separable contacts. Hybrid circuit breakers reduce arcing by using electronics to commutate current when the separable contacts are opened. However, the electronics can only withstand the flow of the commutated current for a short period of time before incurring damage. Thus, achieving ultra-fast motion and/or displacement of the separable contacts during an opening operation is especially important in hybrid circuit breakers, in order to achieve the required gap between the separable contacts (often referred to as a “contact gap”) within an extremely short period of time. Thomson coil actuators are able to achieve such ultra-fast motion and displacement that other conventional switching mechanisms cannot achieve.

(4) A Thomson coil assembly comprises a coiled conductor (i.e. a Thomson coil) and a conductive plate positioned in close proximity to the Thomson coil. The Thomson coil is configured to receive current, and when current is supplied to the Thomson coil, the magnetic field generated by the flow of current through the coil exerts a repulsion force on the conductive plate, driving the conductive plate away from the Thomson coil. When Thomson coil actuators are used in circuit interrupters, the conductive plate is typically coupled to the movable conductor assembly. When the trip unit detects a fault condition, the trip unit transmits a signal that causes current to be supplied to the Thomson coil, and the magnetic field produced by the Thomson coil acts upon the conductive plate to drive the movable conductor assembly away from the stationary contact.

(5) The Thomson coil is typically covered with an insulation layer in order to provide insulation between the Thomson coil and the conductive plate, in order to more precisely control the magnitude and orientation of the magnetic fields produced by the Thomson coil. The effectiveness of a Thomson coil actuator, particularly when used as a switching mechanism in a circuit breaker, relies on the manufacturing process used for producing the insulation layer being able to closely and consistently adhere to design specifications. Known methods for large-scale manufacturing of the insulation layer for Thomson coils include using a potting process. The potting process is intended to be carried out by holding the Thomson coil in a fixed position while the insulation material is applied to the Thomson coil, so that a thin insulation layer will form on the Thomson coil. However, known potting processes often result in the insulation layer having non-uniform thickness and the coil windings being uneven due to the unraveling tendency of the conductors used to form Thomson coils. Such deviations from design specifications have a significant effect on the performance of Thomson coil actuators.

(6) There is thus room for improvement in the systems and methods used to produce the insulation layer of Thomson coils.

SUMMARY OF THE INVENTION

(7) These needs, and others, are met by embodiments of an assembly for manufacturing the insulation layer of a Thomson coil disclosed herein. The disclosed assembly embodiments include a base plate, a coil housing, and a cover plate. The base plate securely seats the coil housing, and the coil housing securely seats a Thomson coil and the cover plate. The cover plate has several ribs that hold multiple turns of a Thomson coil in place while epoxy is applied to the coil, thus ensuring that the epoxy is evenly distributed on the coil surface and that the coil windings remain level. The cover plate and coil housing are structured to either receive a high-pressure epoxy injection or to be used in an epoxy potting process, during which all exposed areas of the Thomson coil are coated by liquid epoxy. After the epoxy has solidified, the Thomson coil is coupled to the coil housing, and the housed and insulated Thomson coil can be removed from the assembly. Any areas of the

Thomson coil that were engaged by the ribs during the application of the epoxy can be overmolded with epoxy after the housed and insulated Thomson coil is removed from the assembly.
(8) In one exemplary embodiment of the disclosed concept, an insulation manufacturing assembly for manufacturing an insulation layer for a Thomson coil comprises: a base plate, a coil housing structured to seat the Thomson coil, a cover plate comprising a plurality of ribs and a plurality of flow passages, and a number of fasteners that fasten the cover plate, the coil housing, and the base plate to one another. The base plate seats the coil housing and the coil housing seats the cover plate. The ribs are structured to engage multiple turns of the Thomson coil and secure the multiple turns in place such that the multiple turns form a level surface. The flow passages are structured to enable a liquid epoxy to flow around the ribs and cover a top surface of the Thomson coil that faces away from the coil housing
(9) In another exemplary embodiment of the disclosed concept, an insulation manufacturing assembly for manufacturing an insulation layer for a Thomson coil comprises: a base plate, a coil housing structured to seat the Thomson coil; a cover plate, and a number of fasteners that fasten the cover plate, the coil housing, and the base plate to one another. The coil housing includes: a coil seating surface with a central opening and a trough disposed adjacent to the central opening. The cover plate includes: a plurality of ribs, a plurality of flow passages, and an overflow cutout. The base plate seats the coil housing and the coil housing seats the cover plate. The ribs are structured to engage multiple turns of the Thomson coil and secure the multiple turns in place such that the multiple turns form a level surface. The flow passages are structured to enable a liquid epoxy to flow around the ribs and cover a top surface of the Thomson coil that faces away from the coil housing. The overflow cutout aligns with the trough such that an excess amount of the liquid epoxy can flow out of the insulation manufacturing assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A full understanding of the invention can be gained from the following description of the preferred embodiments when read in conjunction with the accompanying drawings in which:
(2) FIG. **1**A is an exploded view of an insulation manufacturing assembly for producing an insulation layer for a Thomson coil using a high-pressure injection process, in accordance with an example embodiment of the disclosed concept;
(3) FIG. **1**B is a perspective view of the insulation manufacturing assembly shown in FIG. **1**A;
(4) FIG. **1**C is a sectional view of the insulation manufacturing assembly shown in FIG. **1**B, taken along the line S**1**-S**1** shown in FIG. **1**B and FIG. **5**;
(5) FIG. **1**D is an enlargement of a portion of the sectional view shown in FIG. **1**C, in order to better show details of the ribs of the cover plate shown in FIG. **1**C;
(6) FIG. **2**A is a perspective view of the top side of the coil housing shown in FIGS. **1**A-**1**C;
(7) FIG. **2**B is a perspective view of the bottom side of the coil housing shown in FIGS. **1**A-**1**C;
(8) FIG. **3**A is a perspective view of the top side of the Thomson coil shown in FIGS. **1**A-**1**C;
(9) FIG. **3**B is a perspective view of the bottom side of the Thomson coil shown in FIGS. **1**A-**1**C;
(10) FIG. **4** is a perspective view of the bottom side of the cover plate shown in FIGS. **1**A-**1**C;
(11) FIG. **5** is a sectional view of the insulation manufacturing assembly shown in FIG. **1**C, taken along the line S**2**-S**2** shown in FIG. **1**C, showing points of contact between ribs of the cover plate and the top side of the Thomson coil;
(12) FIG. **6** is a perspective view of the insulation manufacturing assembly shown in FIG. **1**A, with an alternative embodiment of the cover plate shown in FIG. **1**A;
(13) FIG. **7** is an exploded view of an insulation manufacturing assembly for producing an insulation layer for a Thomson coil using a potting process, in accordance with another example

embodiment of the disclosed concept; and
(14) FIG. **8** is a perspective view of the insulation manufacturing assembly shown in FIG. **7**.
DETAILED DESCRIPTION OF THE INVENTION
(15) Directional phrases used herein, such as, for example, left, right, front, back, top, bottom and derivatives thereof, relate to the orientation of the elements shown in the drawings and are not limiting upon the claims unless expressly recited therein.
(16) As employed herein, the statement that two or more parts or components are "coupled" shall mean that the parts are joined or operate together either directly or indirectly, i.e., through one or more intermediate parts or components, so long as a link occurs. As used herein, "directly coupled" means that two elements are directly in contact with each other. As used herein, "fixedly coupled" or "fixed" means that two components are coupled so as to move as one while maintaining a constant orientation relative to each other.
(17) As employed herein, when ordinal terms such as "first" and "second" are used to modify a noun, such use is simply intended to distinguish one item from another, and is not intended to require a sequential order unless specifically stated.
(18) As employed herein, the term "number" shall mean one or an integer greater than one (i.e., a plurality).
(19) Described herein are embodiments of an advantageously designed insulation manufacturing assembly for use in producing the insulation layer of a Thomson coil. The disclosed insulation manufacturing assembly embodiments are structured to hold a Thomson coil securely in place during manufacturing of the insulation layer, thus ensuring that the insulating material is evenly distributed on the surface of the Thomson coil, and that the coil windings will be even and level. It will be appreciated that the disclosed insulation manufacturing assembly embodiments facilitate better large-scale and automated manufacturing of insulation layers for Thomson coil assemblies.
(20) FIGS. **1**A-**1**C respectively show an exploded view, a perspective view, and a sectional view of an insulation manufacturing assembly **100** in accordance with a first embodiment of the disclosed concept, the insulation manufacturing assembly **100** being structured for use in producing an insulation layer for a Thomson coil **10** using a high-pressure injection process. FIG. **1**D shows an enlarged portion of FIG. **1**C in order to show certain details of the insulation manufacturing assembly **100** more clearly. For the sake of brevity, the insulation manufacturing assembly **100** is referred to hereinafter as the "assembly **100**".
(21) For clarity and ease of explanation, the assembly **100**, each of its components, and all other components discussed in relation to the assembly **100** are referred to herein as having a top side and a bottom side. Orientation toward the top side of a component is denoted by the arrow **1** in FIG. **1**A, and orientation toward the bottom side of a component is denoted by the arrow **2** in FIG. **1**B. For example, in accordance with the directions of "top" and "bottom" as denoted by arrows **1** and **2** in FIG. **1**A, only the top side of the components is visible in FIG. **1**A. Similarly, if a first component is positioned nearer to the top side of a second component rather than to the bottom side, then the terms "above" or "upward" can be used to describe the disposition of the first component relative to the second component. Conversely, if a first component is positioned nearer to the bottom side of a second component rather than to the top side, then the terms "below" or "downward" can be used to describe the disposition of the first component relative to the second component. In addition, the assembly **100** comprises a central axis **3** (labeled in FIGS. **1**A and **1**C). Movement or orientation away from the central axis **3** can be described as "lateral", which is indicated by the arrows **4** in FIG. **1**A, while movement or orientation toward the central axis **3** can be described as "medial", which is indicated by the arrows **5** in FIG. **1**A.
(22) The assembly **100** includes a cover plate **102** and a base plate **104** structured to be fixedly coupled together by a number of fasteners **106**, and a coil housing **108** structured to be seated on the top side of the base plate **104** and beneath the cover plate **102**. The top side of the coil housing **108** is structured to seat the Thomson coil **10**. The cover plate **102** comprises a plurality of

apertures **103** and the base plate **104** comprises a plurality of apertures **105**, with the apertures **103** and **105** being structured to receive the fasteners **106**. The fasteners **106** can comprise, for example and without limitation, clamping fasteners, but it should be noted that the fasteners **106** can comprise any type of fastener suitable for maintaining the components of the assembly **100** in a fixed orientation relative to one another during the manufacturing process without departing from the scope of the disclosed concept. Each of the fasteners **106** shown in FIGS. **1**A-**1**B comprises a bolt **106**A and a nut **106**B.
(23) The components of the assembly **100** are advantageously designed to ensure that all components used during the process of producing the insulation layer for the Thomson coil **10** are correctly oriented and coupled to one another during the manufacturing process, i.e. such that the components of the assembly **100** remain in a fixed orientation relative to one another. In particular and as detailed further later herein, the top side of the base plate **104** is structured to receive the bottom side of the coil housing **108** in order to snugly seat the coil housing **108**, the top side of the coil housing **108** is structured to receive the bottom side of the Thomson coil **10** in order to snugly seat the Thomson coil **10**, and a bottom side of the cover plate **102** is structured to engage the top side of the coil housing **108** and surround the top side of the Thomson coil **10** in order to ensure that the Thomson coil **10** remains secured within the coil housing **108** during manufacturing of the insulation layer. After the base plate **104**, coil housing **108**, Thomson coil **10**, and cover plate **102** are seated as described above, the fasteners **106** are inserted through the apertures **103** and **105** and suitably secured to ensure that the components of the assembly **100** remain fixedly coupled to one another.
(24) It is noted that, due to the Thomson coil **10** being circular, the coil housing **108** is consequently circular. Thus, several features of the coil housing **108** detailed later herein are described and referred to in terms of attributes inherent to a circle, including but not limited to “diameter”, “circumference”, etc. However, to the extent that it may be desired to produce an insulation layer for a Thomson coil having another planar, non-circular shape, it will be apparent from the detailed description of the disclosed concepts that the designs disclosed herein for a circular Thomson coil **10** and correspondingly circular coil housing **108** can easily be adapted for other planar shapes, without departing from the scope of the disclosed concept.
(25) As can be seen in FIGS. **1**A and **1**C, the top side of the base plate **104** is formed with a depression **111**, and as can be seen in FIG. **1**C and FIG. **2**B, the bottom side of the coil housing **108** is formed with a protrusion **112** formed in the same shape as the depression **111** and structured to fit snugly within the depression **111**, enabling the base plate **104** to snugly seat the coil housing **108**. As can be seen in FIG. **1**A, FIG. **3**A (showing the top side of the Thomson coil **10**), and FIG. **3**B (showing the bottom side of the Thomson coil **10**), the Thomson coil **10** comprises a coiled planar portion **11** and two leads **12** (one lead being numbered as **12**A and the other lead being numbered as **12**B in FIG. **1**A) extending laterally away from the center of the planar portion **11**, with each lead **12** being one end of the conductor used to form the planar portion **11**. The lead **12**A can be thought of as the start of the outermost turn of the Thomson coil **10**, and as the conductor is wound into increasingly smaller concentric rings to form the planar portion **11**, the lead **12**B routed from the center of the planar portion **11** to be positioned adjacent to and below the bottom side of the planar portion **11** and to extend laterally beyond the outer edge of the planar portion **11**. It will be appreciated that the leads **12** are proportioned to be sufficiently long in order to be connected to a current source, such as when the Thomson coil **10** is placed in operation in a circuit interrupter, for example and without limitation.
(26) As can be seen in FIG. **1**A and FIG. **2**A, the top side of the coil housing **108** comprises a central opening **113** and a trough **114**. The trough **114** extends between the central opening **113** and a lateral edge of the coil housing **108**, such that one end of the trough **114** is adjacent to the central opening **113**, and a bottom surface of the trough **114** is disposed above the bottom side of the coil housing **108** and below a flat coil seating surface **115** of the coil housing **108**. The trough **114** is

thus structured and positioned to receive the two leads **12** of the Thomson coil **10** and to enable the bottom side of the coil planar portion **11** to lie flat upon the coil seating surface **115** so that the coil **10** can be seated on the top side of the coil housing **108**.
(27) As shown in FIG. **2**A, the coil housing **108** includes a flange **116** that extends upward from the coil seating surface **115**, thereby forming a lip **117** that surrounds the flange **116**, with the outer circumference of the flange **116** coinciding with the inner circumference of the lip **117**. In the views shown in FIGS. **2**A and **1**D, it can be seen that the top surface of the lip **117** is co-planar with the coil seating surface **115**. The flange **116** is proportioned to fit around the circumference of the Thomson coil **10**. The flange **116** is also formed with a number of alignment grooves **118** (FIG. **2**A) that are structured to receive corresponding alignment tabs of the cover plate **102** (detailed later herein). Each alignment groove **118** is formed as a cutout in the top of the flange **116**, such that a bottom surface of the alignment groove **118** is disposed above the lip **117** and the coil seating surface **115**, and such that the alignment groove **118** extends between the outer diameter and the inner diameter of the flange **116**.
(28) Referring now to FIG. **4**, the bottom side of the cover plate **102** is formed with a circular depression **120** that extends upward relative to a bottom flat surface **121** of the cover plate **102** such that a circular surface **122** of the circular depression **120** is disposed above the bottom flat surface **121** (it is noted that the term "upward" is used in accordance with the directions of "top" and "bottom" denoted by arrows **1** and **2** in FIG. **1**A). For reasons that will become apparent later herein, the circular surface **122** is referred to hereinafter as the "flow passage surface **122**". The circular depression **120** is structured to receive the coil housing flange **116** such that the flow passage surface **122** engages the top surface of the flange **116** when the cover plate **102** is seated on the coil housing **108**, as shown in FIG. **1**D. The upward extension of the circular depression **120** relative to the bottom flat surface **121** results in the flow passage surface **122** being disposed above the Thomson coil **10** in the assembly **100**, as shown in FIG. **1**D. Still referring to FIG. **1**D, a chamber **123** is formed when the cover plate **102** is seated on the coil housing **108** due to the engagement between the flow passage surface **122** and the flange **116**. The coil seating surface **115** of the coil housing **108** forms the bottom of the chamber **123**, and the flow passage surface **122** of the cover plate **102** forms the top of the chamber **123**. In the medial to lateral dimension, the chamber **123** extends between a downward protruding rim **128** (detailed later herein) of the cover plate **102** and the flange **116** of the coil housing **108**.
(29) Referring once more to FIG. **4**, the bottom side of the cover plate **102** is also formed with a number of alignment tabs **124**, with each alignment tab **124** being structured to be inserted into a corresponding one of the alignment grooves **118** (FIG. **2**A) of the coil housing **108**. While the cover plate **102** and the coil housing **108** are respectively depicted in the figures as having two alignment tabs **124** and two alignment grooves **118**, it will be appreciated that these components can be formed with fewer or more than two alignment tabs **124** and alignment grooves **118** without departing from the scope of the disclosed concept. Inserting the alignment tabs **124** into the alignment grooves **118** prevents the cover plate **102** and coil housing **103** from moving laterally relative to one another.
(30) In addition to the features noted above, each of the base plate **104**, coil housing **108**, and cover plate **102** comprise features aligning with the central axis **3** of the assembly **100** in order to ensure that all three of these components are properly centered when the assembly **100** is assembled. These features are especially apparent in the sectional view shown in FIG. **1**C. Specifically, the base plate **104** comprises a central opening **125** and an upward protruding rim **126** that surrounds the central opening **125** (see FIGS. **1**A and **1**C), the cover plate **102** comprises a central opening **127** and a downward protruding rim **128** that surrounds the central opening **127** (see FIGS. **1**A, **1**C, and **4**), and the coil housing **108** comprises the central opening **113** (see FIGS. **1**C and **2**A-**2**B) structured to receive the upward protruding rim **126** and the downward protruding rim **128**. The downward protruding rim **128** is also structured to be received by a central opening **14** of the

Thomson coil **10**.
(31) The upward protruding rim **126** and downward protruding rim **128** are structured to function as guide portions that respectively guide the coil housing **108** to be properly seated on the base plate **104** and guide the cover plate **102** to be properly seated on the coil housing **108**. The upward protruding rim **126** and the downward protruding rim **128** are also structured to receive one of the fasteners **106**, which can be referred to as the central fastener **106′** (due to being aligned with the central axis **3** when inserted into the upward and downward protruding rims **126**, **128**. After the central fastener **106′** is inserted into the upward and downward protruding rims **126**, **128**, it can be suitably secured in order to couple the base plate **104**, the coil housing **108**, the Thomson coil **10**, and the cover plate **102** to one another and maintain the central openings **127**, **14**, **113**, **125** in alignment with the central axis **3** during the insulation layer manufacturing process. In addition, it is noted that the downward protruding rim **128** of the cover plate **102** is structured to fit snugly within the coil housing central opening **113**, in order to prevent liquid epoxy from flowing downward below the Thomson coil **10** during the process of producing the insulation layer, as detailed later herein.
(32) Referring once more to FIG. **4** and in conjunction with FIG. **1**D, one of the advantageous features of the assembly **100** is a plurality of ribs **130** formed on the bottom side of the cover plate **102**. As best shown in FIG. **1**D, each rib **130** extends downward relative to the flow passage surface **122** of the circular depression **120**. As shown in FIG. **1**D, when the Thomson coil **10** and the cover plate **102** are seated on the coil housing **108**, the bottom surface of each rib **130** engages the top surface of the Thomson coil **10**, while the flow passage surface **122** of the cover plate **102** remains spaced a distance **131** away from the top surface of the Thomson coil **10** (distance **131** may also be referred to hereinafter as “height **131**”, as context necessitates). The spaces between the ribs **130** (i.e. the spaces of height **131** between the flow passage surface **122** and the Thomson coil **10** in the chamber **123**) form material flow passages **133** (numbered in FIG. **1**D, FIG. **4**, and FIG. **5**) through which injected liquid insulative molding material can flow. Each rib **130** is proportioned and positioned to be able to secure multiple turns **15** of the Thomson coil **10** at a time (the turns **15** being numbered in FIG. **1**D), in order to ensure that the planar portion **11** forms a level surface when the insulation layer is forming, i.e. such that the turns **15** lie flat relative to one another. In addition to keeping the turns **15** of the Thomson coil **10** secure to ensure that the planar portion **11** forms a level surface, the bottom surfaces of the ribs **130** contact a relatively small portion of the surface area of the Thomson coil **10**, thus minimizing the surface area that the injected insulating material cannot cover.
(33) Referring to FIG. **4** in conjunction with FIG. **2**A, it is noted that the cover plate **102** and the coil housing **108** are each formed with a respective material injection cutout **134**, **135**, and that the material injection cutouts **134**, **135** align with one another to form a material injection path **136** when the cover plate **102** is seated on the coil housing **108**. The material injection path **136** is numbered in FIG. **5**, although it is noted that only the bottom portion of the material injection path **136** can be seen in the sectional view of FIG. **5**. The material injection path **136** forms a tunnel leading from the exterior of the assembly **100** to the interior of the chamber **123** (the chamber being numbered in FIG. **1**D), and specifically leads to one of the material flow passages **133** of the chamber **123**. Thus, the material injection path **136** is in fluid communication with the material flow passages **133**, and is structured to receive a high-pressure injector that can inject liquid insulative molding material (for example and without limitation, liquid epoxy) into the chamber **123** in order to form the insulation layer of the Thomson coil **10**. For the sake of brevity, the liquid insulative molding material will be referred to hereinafter as being liquid epoxy, but it should be understood that liquid materials other than epoxy can be used without departing from the scope of the disclosed concept, provided that said materials can harden into a solid and have sufficient insulation capability for use with a Thomson coil **10**.
(34) In FIG. **1**D and FIG. **5**, it can be seen that there is space **141** between the outer edge of the

Thomson coil **10** and the inner circumference of the coil housing flange **116**. This space **141** is referred to hereinafter as the "material flow channel **141**". It is noted that the channel **141** varies in width in the lateral dimension. In addition, there is a gap **143** between all adjacent turns **15** of the Thomson coil **10** (see FIG. **1**D), each gap **143** being referred to hereinafter as a "coil turn gap **143**". When liquid epoxy is injected into the chamber **123** through the material injection path **136**, the liquid epoxy flows over the top surface of the Thomson coil **10** through the material flow passages **133**, into the material flow channel **141**, and into the coil turn gaps **143**. The epoxy that fills the material flow channel **141** serves to keep the Thomson coil **10** fixed in place within the coil housing **108**.

(35) The cover plate **102** is also formed with an overflow cutout **145** (FIG. **4**) that is positioned to align with at least a lateral end of the trough **114** (FIG. **2**A) of the coil housing **108** in order to form an overflow channel **147** (FIG. **5**) when the cover plate **102** is seated on the coil housing **108**. The overflow channel **145** enables the interior of the chamber **123** (FIG. **1**D) to be in fluid communication with the exterior of the assembly **100**. The chamber **123** is structured such that, after the epoxy has been injected into the chamber **123** and has filled all of the material flow passages **133**, the material flow channel **141**, and the coil turn gaps **143**, any excess epoxy will flow out of the chamber **123** through the overflow channel **147** to the exterior of the assembly **100**.

(36) Once the epoxy injection is complete and the liquid epoxy has solidified, the manufacturing of the insulation layer is complete. The final product of the insulation manufacturing process using the assembly **100** is a unitary structure comprising the Thomson coil **10** seated in and coupled to the coil housing **108** via the epoxy, with a layer of solidified epoxy coating the Thomson coil **10**, and this final product is referred to hereinafter as the "housed and insulated Thomson coil **10**′" (not shown in the figures). After the epoxy has solidified, the fasteners **106** are removed from the assembly **100**, the cover plate **102** is unseated from the top of the coil housing **108**, and the coil housing **108** is unseated from the base plate **104**.

(37) Prior to coupling the components of the assembly **100** together and injecting the epoxy into the material injection path **136**, anti-adhesion primer is applied to the surfaces of the cover plate **102** to facilitate easy removal of the cover plate **102** from the epoxy-coated Thomson coil **10** and coil housing **108** after the manufacturing process is complete. In addition, the downward protruding rim **128** of the cover plate **102** prevents the epoxy from flowing into the coil central opening **14** and the central opening **113** of the coil housing **108**, in order to facilitate easy removal of the cover plate **102** and base plate **104** after the manufacturing process is complete.

(38) If needed, any areas of the housed and insulated Thomson coil **10**′ that are not covered by epoxy (i.e. those areas of the top side of the Thomson coil **10** that were engaged by the ribs **130** of the cover plate **102**) can be filled in with insulating material using any suitable method, for example and without limitation, by overmolding, or using insulating tape or insulating plastic parts. However, in certain instances, it may not be necessary to fill in the non-covered areas, depending on the type of wire conductor used to form the Thomson coil. For example, wires typically include film insulation and many wires include an insulating finish such as varnish or shellac around the conductive portion of the wire, which may eliminate the need to fill in the areas of the Thomson coil **10** that were engaged by the ribs **130**. However, shellac in particular can crack, so it may be desirable to fill in the gaps on the top surface of the Thomson coil **10** with epoxy even if the wire includes an insulating finish.

(39) It should be noted that the location of the material injection path **136** shown in FIG. **5** is representative and that a material injection path can be formed elsewhere in the assembly **100** without departing from the scope of the disclosed concept. In one non-limiting illustrative example shown in FIG. **6**, the assembly **100** includes a cover plate **102**′ instead of the cover plate **102**, and includes a coil housing **108**′ instead of the coil housing **108**. (It is noted that the coil housing **108**′ is identical to the coil housing **208** shown in FIG. **7**, as part of an assembly **200** described later herein.) The cover plate **102**′ does not include a material injection cutout **134** and the coil housing

108′ does not include a material injection cutout **135**, due to the cover plate **102′** instead comprising a thru-hole **137** that extends from the top side of the cover plate **102′** to the bottom side of the cover plate **102′**. The thru-hole **137** is an alternative embodiment of the material injection path **136** and enables a high-pressure epoxy injector to deposit the epoxy directly into one of the material flow passages **133** from a position located above the top side of the Thomson coil **10**. This manufacturing method allows precise control of the epoxy insulation layer thickness on top of the Thomson coil **10**, which is a significant factor in the performance of the Thomson coil **10**. It is noted that the thru-hole is not drawn to scale in FIG. **6** and is depicted as being larger than its actual size, in order to depict the access to the top side of the Thomson coil **10** afforded by the thru-hole **137**.

(40) Referring now to FIGS. **7-8**, an insulation manufacturing assembly **200** in accordance with another embodiment of the disclosed concept is shown, the insulation manufacturing assembly **200** being structured for use in producing an insulation layer for a Thomson coil **10** using a potting process. The insulation manufacturing assembly **200** is referred to hereinafter as the "assembly **200**" for brevity. The assembly **200** is identical to the assembly **100** in several respects, as it is only the cover plate **202** of the assembly **200** that is notably distinct from the cover plate **102** of the assembly **100**. Accordingly, those features of the assembly **200** that structurally and functionally correspond to features of the assembly **100** are numbered in FIGS. **7-8** with the same reference numbers used in FIGS. **1**A-**6**, but incremented by 100. For example and without limitation, the assembly **200** includes a base plate **202** and fasteners **206** that are identical to the base plate **102** and fasteners **106** of assembly **100**. The coil housing **208** is identical to the coil housing **108**, except that the coil housing **208** does not include a material injection cutout. (As previously noted in connection with the description of the cover plate **102′** and coil housing **108′** shown in FIG. **6**, the coil housing **108′** is identical to the coil housing **208** shown in FIG. **7**).

(41) In addition, the components of assembly **200** are structured to be coupled together in the same manner as the components of the assembly **100** prior to commencement of the insulation layer manufacturing process. It is noted that the assembly **200** is structured to produce an insulation layer for the same type of Thomson coil **10** as the assembly **100**. For the sake of brevity, those features of the assembly **200** that are structurally and functionally identical to those of the assembly **100** are not described again hereinafter. Only the features of the cover plate **202** will be described in detail, and it should be understood that all other components of the assembly **200** are structurally and functionally identical to their correspondingly numbered component of the assembly **100**.

(42) Reference now made to FIG. **8**. In contrast with assembly **100**, assembly **200** is structured to be used for manufacturing the insulation layer of the Thomson coil **10** using a potting process rather than a high-pressure injection process. Accordingly, the components of the assembly **200** do not include features corresponding to the material injection cutouts **134**, **135** and material injection path **136** of the assembly **100**. Instead, the cover plate **202** includes several large openings **251**, with each opening **251** functioning as a potting section into which epoxy can be poured. An opening **251** may be referred to hereinafter as a "potting section **251**", as context necessitates.

(43) Although shaped and positioned differently from the ribs **130** of the cover plate **102**, the cover plate **202** also includes a plurality of ribs **230**, which includes both a plurality of partitioning ribs **230**A and partial ribs **230**B (detailed further later herein). When the Thomson coil **10** and the cover plate **202** are seated on the coil housing **208**, the bottom surface of each rib **230** engages the top surface of the Thomson coil **10** and is proportioned and positioned to be able to secure multiple turns **15** of the Thomson coil **10** at a time, in order to ensure that the Thomson coil **10** forms a level surface when the epoxy layer is forming, i.e. such that the turns **15** lie flat relative to one another. While the cover plate **202** is depicted in the figures as comprising four partitioning ribs **230**A and four partial ribs **230**B, it will be appreciated that greater or fewer than four ribs **230**A and **230**B can be included in the cover plate **202** without departing from the scope of the disclosed concept.

(44) The partitioning ribs **230**A extend from a lateral border **252** of the cover plate **202** to a

downward protruding rim **228** (aligned with the central axis **3**) that receives the central fastener **206**′, with the partitioning ribs **230**A creating the distinct potting sections **251** such that the interior of each potting section **251** is isolated from the interior of every other potting section **253**. It will be appreciated that the number of potting sections **251** is dependent upon the number of partitioning ribs **230**A.

(45) Within each potting section **251** is a partial rib **230**B that extends from the lateral border **252** toward the downward protruding rim **228** but does not reach the downward protruding rim **228**. For each partial rib **230**B, the end that is not connected to the lateral border **252** is the free end **254**. The gap between each of the free ends **254** and the downward protruding rim **228** serves as a material flow passage **233** that enables epoxy poured into one area of a potting section **251** to flow and fill in the entirety of that potting section **251**. It will be appreciated that more than one partial rib **230**B can be included in each potting section **251** without departing from the scope of the disclosed concept.

(46) Once pouring of the epoxy is complete and the liquid epoxy has solidified, the manufacturing of the insulation layer is complete. The final product of the insulation manufacturing process using the assembly **200** is a unitary structure comprising the Thomson coil **10** seated in and coupled to the coil housing **208** via the epoxy, with a layer of solidified epoxy coating the Thomson coil **10**, and this final product is referred to hereinafter as the “housed and insulated Thomson coil **10**″” (not shown in the figures). After the epoxy has solidified, the fasteners **206** are removed from the assembly **100**, the cover plate **202** is unseated from the top of the coil housing **208**, and the coil housing **208** is unseated from the base plate **204**.

(47) Prior to coupling the components of the assembly **200** together and pouring epoxy into each of the potting sections **251**, anti-adhesion primer is applied to the surfaces of the cover plate **202** to facilitate easy removal of the cover plate **202** from the epoxy-coated Thomson coil **10** and coil housing **208** after the epoxy has set. The assembly **200** is placed into a vacuum chamber after the epoxy is poured in order to eliminate any air bubbles that may form during pouring of the epoxy. It will be appreciated that achieving uniform thickness of the insulation layer is harder to achieve with potting using the assembly **200** than with the injection process performed with the assembly **100**, so after the epoxy has set and the cover plate **202** is removed from the housed and insulated coil **10**″, the potted epoxy layer is machined in order to bring the epoxy layer to uniform thickness. It will be appreciated that sanding, grinding, or any other method suitable for leveling a surface can be used to even out the potted epoxy layer.

(48) Similarly to the housed and insulated coil **10**′, if needed, any areas of the housed and insulated Thomson coil **10**″ that are not covered by epoxy (i.e. those areas of the top side of the Thomson coil **10** that were engaged by the ribs **230** of the cover plate **202**) can be filled in with insulating material using any suitable method, for example and without limitation, by overmolding, or using insulating tape or insulating plastic parts.

(49) While there are obvious differences between the structure of the cover plate **202** and the cover plate **102**, the assembly **100** and the assembly **200** have significant features in common. The ribs **130** and ribs **230** enable epoxy to flow freely through the material flow passages **133** and **233**, while ensuring that all turns **15** of the Thomson coil remain level and minimizing how much exposed surface area of the Thomson coil **10** is prevented from being covered by the epoxy during the injection or potting process. Both the assembly **100** and the assembly **200** facilitate greatly improved large-scale and automated manufacturing of housed and insulated Thomson coils with a uniform insulation layer.

(50) While specific embodiments of the invention have been described in detail, it will be appreciated by those skilled in the art that various modifications and alternatives to those details could be developed in light of the overall teachings of the disclosure. Accordingly, the particular arrangements disclosed are meant to be illustrative only and not limiting as to the scope of

disclosed concept which is to be given the full breadth of the claims appended and any and all equivalents thereof.

Claims

1. An insulation manufacturing assembly for manufacturing an insulation layer for a Thomson coil, the assembly comprising: a base plate; a coil housing structured to seat the Thomson coil; a cover plate comprising a plurality of ribs and a plurality of flow passages; and a number of fasteners that fasten the cover plate, the coil housing, and the base plate to one another, wherein the base plate seats the coil housing and the coil housing seats the cover plate, wherein the ribs are structured to engage multiple turns of the Thomson coil and secure the multiple turns in place such that the multiple turns form a level surface, and wherein the flow passages are structured to enable a liquid epoxy to flow around the ribs and cover a top surface of the Thomson coil that faces away from the coil housing.
2. The insulation manufacturing assembly of claim 1, wherein the coil housing comprises a central opening structured to align with a central opening of the Thomson coil, wherein the base plate comprises an upward protruding rim inserted snugly within the central opening of the coil housing, wherein the cover plate comprises a downward protruding rim inserted snugly within the central opening of the coil housing, and wherein the upward protruding rim and downward protruding rim align with one another such that one fastener of the number of fasteners is inserted through the downward protruding rim, the Thomson coil central opening, the coil housing central opening, and the upward protruding rim.
3. The insulation manufacturing assembly of claim 2, wherein the coil housing is structured to seat a bottom surface of the Thomson coil, the bottom surface of the Thomson coil being disposed opposite the top surface of the Thomson coil, wherein the cover plate comprises a flow passage surface structured to face the top surface of the Thomson coil, wherein the ribs extend from the flow passage surface toward the top surface of the Thomson coil, and wherein the flow passage surface is structured such that the flow passages are disposed between the ribs and there is one continuous space formed comprising the flow passages.
4. The insulation manufacturing assembly of claim 2, wherein the coil housing comprises a top side that faces toward the cover plate and a bottom side disposed opposite the top side that faces the base plate, wherein the top side of the coil housing is formed with a coil seating surface, a flange, and a lip, such that the flange surrounds the coil seating surface and the lip surrounds the flange, wherein the flange is formed with an injection cutout, wherein the injection cutout is structured to receive a high-pressure epoxy injector, and wherein the flow passages are structured such that injected liquid epoxy can flow freely along the top surface of the Thomson coil, flow into any spaces between individual turns of the Thomson coil, and flow into any spaces between an outermost edge of the coil and the coil housing.
5. The insulation manufacturing assembly of claim 2, wherein the cover plate comprises a flow passage surface structured to face the top surface of the Thomson coil, wherein the ribs extend from the flow passage surface toward the top surface of the Thomson coil, wherein the cover plate comprises a thru-hole extending from a top side of the cover plate to a bottom side of the cover plate and disposed between the ribs, and wherein the thru-hole is disposed such that injecting the liquid epoxy into the thru-hole enables the liquid epoxy to flow through the flow passages.
6. The insulation manufacturing assembly of claim 2, wherein the downward protruding rim is structured to prevent the liquid epoxy from flowing into the Thomson coil central opening and the coil housing central opening.
7. The insulation manufacturing assembly of claim 2, wherein the coil housing is structured to seat a bottom surface of the Thomson coil, the bottom surface of the Thomson coil being disposed opposite the top surface of the Thomson coil, wherein the cover plate comprises a plurality of

partitioning ribs and a central portion that includes the downward protruding rim, wherein the partitioning ribs extend from a border of the cover plate to the central portion, wherein the partitioning ribs form a plurality of potting sections such that, an interior of each potting section is isolated from an interior of every other potting section, wherein each potting section comprises at least one partial rib that extends from the border portion but does not reach the central portion, and wherein the cover plate is structured such that the flow passages only allow liquid epoxy poured within a given potting section to flow within that given potting section.

8. The insulation manufacturing assembly of claim 7, wherein the downward protruding rim is structured to prevent the liquid epoxy from flowing into the Thomson coil central opening and coil housing central opening.

9. An insulation manufacturing assembly for manufacturing an insulation layer for a Thomson coil, the assembly comprising: a base plate; a coil housing structured to seat the Thomson coil, the coil housing comprising: a coil seating surface with a central opening; and a trough disposed adjacent to the central opening; a cover plate, the cover plate comprising: a plurality of ribs; a plurality of flow passages; and an overflow cutout; a number of fasteners that fasten the cover plate, the coil housing, and the base plate to one another, wherein the base plate seats the coil housing and the coil housing seats the cover plate, wherein the ribs are structured to engage multiple turns of the Thomson coil and secure the multiple turns in place such that the multiple turns form a level surface, wherein the flow passages are structured to enable a liquid epoxy to flow around the ribs and cover a top surface of the Thomson coil that faces away from the coil housing, and wherein the overflow cutout aligns with the trough such that an excess amount of the liquid epoxy can flow out of the insulation manufacturing assembly.

10. The insulation manufacturing assembly of claim 9, wherein the trough is disposed below the coil seating surface and structured to receive any lead of the Thomson coil that is disposed below a planar portion of the Thomson coil.

11. The insulation manufacturing assembly of claim 9, wherein the cover plate further comprises a bottom flat surface and a flow passage surface disposed above the bottom flat surface, wherein the plurality of ribs extend downward from the flow passage surface such that a bottom surface of each rib is disposed above the bottom flat surface.

12. The insulation manufacturing assembly of claim 11, wherein the overflow cutout is formed in the bottom flat surface.